Practical Demonstration

Candidate Name:	
Assessor Name:	
Qualification:	**Certificate in AI in Immersive Technology, Level – 6**
Assessment Centre:	
Date of Assessment:	

Instructions:

Read and understand the directions carefully:

- this practical demonstration is based on the performance criteria from all or some of the units of competency in AI in Immersive Technology, Level – 6
- this assessment activity will be used to measure your underpinning skills
- you will have fifteen (15) minutes to familiarise yourself with the resources to be used
- you have five (5) hours to complete this demonstration

Job Sheet AI in Immersive Technology, Level – 6

Job 01:	Develop Program Using Python
Time:	40 Minutes
Unit Covered:	OU-ICT-AIIT-01-L6-V1: Apply Math Skills for Machine Learning OU-ICT-AIIT-02-L6-V1: Develop programs using Python

Working Procedure/ Steps:

1. Wear the required Personal Protective Equipment (PPE) for the job.
2. Observe and follow appropriate occupational health and safety (OHS) requirements during the demonstration.
3. Check Electricity, Peripheral device & Internet Connections to your Computer.
4. Start the Computer.
5. Check for software errors and troubleshoot problems.
6. Collect the resources (Dataset) and materials from your assessor as per the job requirement.
7. Set up the Python environment required to complete the project.
8. Load the population dataset into the environment.
9. Perform initial data inspection.
10. Compute the mean, median, variance, and standard deviation of population for each continent.
11. Calculate cumulative population growth over the years for each country.
12. Change the column names of the dataset (e.g., 2020 Population → 2020) using conditional logic.
13. Write the cleaned dataset (with updated column names) into a file named `processed_data.csv`.
14. Visualize continent-wise population in **1900** and **2022** using histograms.
15. Show the timeline of continent-wise population and interpret trends and anomalies using statistical reasoning.
16. Show the correlation between numerical columns by computing the **correlation matrix** and analyze the relationships using statistical reasoning.
17. Submit the final output to the competency assessor.
18. Shut down the computer and clean your workplace.

Specification Sheet for AI in Immersive Technology, Level – 6	
Job 01:	Develop Program Using Python
Condition For the Job:	
Work must be carried out safely and according to AI in Immersive Technology, Level – 6 standards, following the following established best practices for system usage, data handling, and software development.	
Resources Required:	
Tools, Equipment, and furniture:	1. Personal Computer – 1 set 2. Ergonomic Chair – 1 Pc
Software:	1. Operating software – Windows or Linux 2. Programming and ML Tools 2.1 Python (version 3.8 or later) 2.2 Jupyter Notebook / JupyterLab 2.3 Integrated Development Environment (IDE): VS Code / PyCharm
PPE:	1. Rubber sole Shoes/Sandal – 1 pair

Job Sheet for AI in Immersive Technology, Level – 6	
Job 02:	Create Project on Artificial Intelligence (AI) and Machine Learning (ML)
Time:	60 Minutes
Unit Covered:	OU-ICT-AIIT-03-L6-V1: Create Project on AI and ML

Working Procedure/ Steps:

1. Wear the required Personal Protective Equipment (PPE) for the job.
2. Observe and follow appropriate occupational health and safety (OHS) requirements during the demonstration.
3. Check Electricity, Peripheral device & Internet Connections to your Computer.
4. Start the Computer.
5. Check for software errors and troubleshoot problems.
6. Collect the resources (Dataset) and materials from your assessor as per the job requirement.
7. Create a **Weather Prediction** System
8. Set up the Python environment required to create the system.
9. Load the weather dataset into the environment.
10. Perform initial data inspection.
11. Define the ML Problem Statement
12. Perform Feature Selection
13. Split the dataset for training and testing
14. Implement a baseline ML model
15. Apply feature scaling, compare model performance before and after scaling
16. Train other optimized ML models (at least 2 Model)
17. Evaluate Model Performance
18. Visualize Model Performance
19. Detect Overfitting and Underfitting
20. Save and deploy the final model
21. Submit the final output to the competency assessor.
22. Shut down the computer and clean your workplace.
23. Clean tools, equipment, materials, and work area.

Specification Sheet for AI in Immersive Technology, Level – 6	
Job 02:	Create Project on Artificial Intelligence (AI) and Machine Learning (ML)
Condition For the Job:	
Work must be carried out safely and according to AI in Immersive Technology, Level – 6 standards, following the following established best practices for system usage, data handling, and software development.	
Resources Required:	
Tools, Equipment, and furniture:	1. Personal Computer – 1 set 2. Ergonomic Chair – 1 Pc
Software:	1. Operating software – Windows or Linux 2. Programming and ML Tools 2.1 Python (version 3.8 or later) 2.2 Jupyter Notebook / JupyterLab 2.3 Integrated Development Environment (IDE): VS Code / PyCharm
PPE:	1. Rubber sole Shoes/Sandal – 1 pair

Job Sheet for AI in Immersive Technology, Level – 6	
Job 03:	Implement of Deep Learning
Time:	60 Minutes
Unit Covered:	OU-ICT-AIIT-04-L6-V1: Implement of Deep Learning

Working Procedure/ Steps:

1. Wear the required Personal Protective Equipment (PPE) for the job.
2. Observe and follow appropriate occupational health and safety (OHS) requirements during the demonstration.
3. Check Electricity, Peripheral device & Internet Connections to your Computer.
4. Start the Computer.
5. Check for software errors and troubleshoot problems.
6. Collect the required resources (dataset and materials) from the assessor as per the job requirement.
7. Create a **Sentiment Analysis** System using Deep Learning
8. Set up the Python environment required to complete the project.
9. Load and inspect the dataset.
10. Prepare the data (cleaning, noise removal, and data annotation).
11. Extract relevant features and transform the data (e.g., tokenization, embedding, normalization).
12. Split the dataset into training, validation, and testing sets. (added – mandatory for DL)
13. Design the deep learning model architecture.
14. Compile and train the model.
15. Evaluate the model performance using appropriate metrics.
16. Visualize model performance (loss, accuracy, or error trends).
17. Detect overfitting and underfitting using training and validation results.
18. Save and deploy the final trained model.
19. Submit the final output to the competency assessor.
20. Shut down the computer and clean your workplace.
21. Clean tools, equipment, materials, and work area.

Specification Sheet for AI in Immersive Technology, Level – 6	
Job 03:	Implement of Deep Learning
Condition For the Job:	
Work must be carried out safely and according to AI in Immersive Technology, Level – 6 standards, following the following established best practices for system usage, data handling, and software development.	
Resources Required:	
Tools, Equipment, and furniture:	1. Personal Computer – 1 set 2. Ergonomic Chair – 1 Pc
Software:	1. Operating software – Windows or Linux 2. Programming and ML Tools 2.1 Python (version 3.8 or later) 2.2 Jupyter Notebook / JupyterLab 2.3 Integrated Development Environment (IDE): VS Code / PyCharm
PPE:	2. Rubber sole Shoes/Sandal – 1 pair

Job Sheet for AI in Immersive Technology, Level – 6	
Job 04:	Work with Immersive Technology
Time:	70 Minutes
Unit Covered:	OU-ICT-AIIT-04-L6-V1: Work with Immersive Technology

Working Procedure/ Steps:

1. Wear the required Personal Protective Equipment (PPE) for the job (if applicable).
2. Observe and follow appropriate Occupational Health and Safety (OHS) requirements during the demonstration.
3. Construct an Augmented Reality Application
4. Check the AR device, power supply, and peripheral connections (e.g., phone/tablet, headset, controller).
5. Start the AR platform/device and ensure it powers up correctly.
6. Open the Unity AR project on the system.
7. Check for software or device errors and troubleshoot any issues.
8. Collect any required 3D assets or datasets from the assessor as per the project requirements.
9. Ensure the AR tracking system is enabled and calibrated (camera tracking, surface detection).
10. Load the pre-developed AR application into the environment.
11. Perform an initial inspection of the AR scene, confirming that all objects are loaded correctly.
12. Identify and verify interactive objects within the AR environment.
13. Demonstrate surface detection by pointing the device to a real-world plane (floor, table, or flat surface).
14. Place virtual 3D objects onto the detected surface in the AR environment.
15. Perform at least one user interaction with the virtual object (e.g., tap, rotate, scale).
16. Observe and demonstrate system response to the interaction in real time.
17. Explain the scene structure, tracking logic, and interaction implementation to the assessor.
18. Identify at least one user safety or comfort consideration (e.g., motion sickness, eye strain).
19. Identify at least one limitation of the AR prototype (e.g., lighting conditions, object scale constraints).
20. Demonstrate the correct shutdown procedure for the AR application and device.
21. Save any logs, screenshots, or results required for the assessment.
22. Clean the AR device, peripherals, and workstation

Specification Sheet for AI in Immersive Technology, Level – 6	
Job 04:	AR Project on Immersive Technology
Condition For the Job:	
Work must be carried out safely and according to AI in Immersive Technology, Level – 6 standards, following the following established best practices for system usage, data handling, and software development.	
Resources Required:	
Tools, Equipment, and furniture:	1. Personal Computer – 1 set 2. Ergonomic Chair – 1 Pc
Software:	1. Operating software – Windows 2. Setup environment required for AR, VR
Hardwar:	1. AR Device (Android / IOS) 2. CCTV Camera/ IP Camera 3. Workstation for Immersive Application development 4. Hand kit
PPE:	1. Rubber sole Shoes/Sandal – 1 pair

Job Sheet for AI in Immersive Technology, Level – 6	
Job 05:	VR Project on Immersive Technology
Time:	70 Minutes
Unit Covered:	OU-ICT-AIIT-04-L6-V1: Work with Immersive Technology

Working Procedure/ Steps:

1. Wear the required Personal Protective Equipment (PPE) for the job (if applicable).
2. Observe and follow appropriate Occupational Health and Safety (OHS) requirements during the demonstration.
3. Create a Virtual Reality Application
4. Check the VR device, power supply, and peripheral connections (e.g., phone/tablet, headset, controller).
5. Start the VR platform/device and ensure it powers up correctly.
6. Open the Unity VR project on the system.
7. Check for software or device errors and troubleshoot any issues.
8. Collect any required 3D assets or datasets from the assessor as per the project requirements.
9. Ensure the VR tracking system is enabled and calibrated (camera tracking, surface detection).
10. Load the pre-developed VR application into the environment.
11. Perform an initial inspection of the VR scene, confirming that all objects are loaded correctly.
12. Identify and verify interactive objects within the VR environment.
13. Demonstrate surface detection by pointing the device to a real-world plane (floor, table, or flat surface).
14. Place virtual 3D objects onto the detected surface in the VR environment.
15. Perform at least one user interaction with the virtual object (e.g., tap, rotate, scale).
16. Observe and demonstrate system response to the interaction in real time.
17. Explain the scene structure, tracking logic, and interaction implementation to the assessor.
18. Identify at least one user safety or comfort consideration (e.g., motion sickness, eye strain).
19. Identify at least one limitation of the VR prototype (e.g., lighting conditions, object scale constraints).
20. Demonstrate the correct shutdown procedure for the VR application and device.
21. Save any logs, screenshots, or results required for the assessment.
22. Clean the VR device, peripherals, and workstation

Specification Sheet for AI in Immersive Technology, Level – 6	
Job 05:	VR Project on Immersive Technology
Condition For the Job:	
Work must be carried out safely and according to AI in Immersive Technology, Level – 6 standards, following the following established best practices for system usage, data handling, and software development.	
Resources Required:	
Tools, Equipment, and furniture:	1. Personal Computer – 1 set 2. Ergonomic Chair – 1 Pc
Hardwar	1. VR set 2. Hand Kit 3. Workstation for Immersive application development 4. CCT Camera/ IP Camera
Software:	1. Operating software – Windows 2. Setup environment required for AR, VR development
PPE:	1. Rubber sole Shoes/Sandal – 1 pair